Syed Izzat Ullah

ML & Robotics

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Syed Izzat Ullah | Corpus Christi, TX, USA

Professional Summary

Ph.D. candidate in Robotics & ML with expertise in learning-based control policies, robot dynamics, uncertainty-aware planning, and real-hardware validation. Experienced in designing and training deep RL and imitation learning architectures for contact-rich and dynamic environments, integrating classical control (MPC, impedance, PID) with modern data-driven methods. Proven track record of transitioning research prototypes to hardware-validated systems, with publications in IEEE and Springer. Seeking to apply policy learning and physical reasoning skills to advance robotic manipulation at scale.

Education

Ph.D. Computer Science — Robotics & ML | 3.75 (4.00) CGPA
Texas A&M University-Corpus Christi, USA | *May 2022– July 2026*

Dissertation focus: "Uncertainty-aware probabilistic navigation for UAVs operating alongside dynamic obstacles, integrating learning-based forecasting with optimization-based planning for robust real-world deployment."

M.S. Electrical Engineering — Robotics & Control | 3.16 (4.00) CGPA
Lahore University of Management Sciences (LUMS), Pakistan | *2017-19*

B.S. Telecommunication Engineering | 3.83 (4.00) CGPA
Balochistan University of IT, Engineering & Management Sciences, Pakistan | *2012-16*

Technical Expertise

Languages:	Python (PyTorch, NumPy, Pandas, SciPy, scikit-learn), C++, MATLAB, Shell, CMake
Robot Learning:	Reinforcement Learning (PPO, SAC, DQN, RLHF, GRPO), Offline RL, Inverse RL, Imitation & Representation Learning
AI/ML:	Deep Learning, Probabilistic Modeling & Uncertainty Quantification, Transformer Architectures, Graph Neural Networks (GNNs), Time-Series & Sequential Modeling, Mixture Density Networks, Classical Machine Learning (Logistic Regression, SVMs, XGBoost, LightGBM), Feature Engineering, Model Training & Evaluation, End-to-End ML Pipelines, Scalable Data Pipelines, Model Optimization, Computer Vision
Control & Dynamics:	MPC, PID, Impedance Control, State Estimation, Dynamics Modeling, Trajectory Opt
Sim-to-Real:	ROS, Gazebo, Unreal Engine + AirSim, CARLA, Sim-to-Real Transfer, Hardware Val
Planning:	Optimization-based Planning, Sampling-based (RRT*, PRM), Uncertainty-aware Nav
Perception & Sensing:	SLAM, Sensor Fusion, IMUs, LiDAR, RGB-D and Stereo Cameras, GPS
Hardware Platforms:	Crazyflie drones, TurtleBot3, Leo Rover, UR5 Manipulator, VICON
Tools & Infrastructure:	Docker, Git, Linux, HPC, SolidWorks, Origin- Lab, LaTex, and CI/CD Pipelines.

Research Experience

Graduate Research Assistant — Texas A&M University-Corpus Christi | May 2022 – Present

- **Designed and developed a transformer-based multi-modal forecasting system** using PyTorch that predicts long-horizon trajectories of non-cooperative dynamic agents, advancing state-of-the-art in temporal sequence prediction for autonomous systems. Implemented CI/CD pipelines for ML model training and deployment (Video).
- **Developed POF+MADER**, an uncertainty-aware trajectory planning framework integrating a probabilistic obstacle filter with optimization-based control, reducing collisions by 39% in simulation (ROS/Gazebo) and 25% in hardware trials on Crazyflie quadrotors (Video).
- **Developed NanoBench**, a multi-task benchmark dataset covering system identification, controller benchmarking, and state estimation for nano-quadrotors using real hardware (Crazyflie 2.1 + VICON mocap), enabling reproducible evaluation of learning-based and classical control methods.
- **Designed a coaxial modular aerial system** with reconfigurable morphology for diverse mission profiles including payload transport, inspection, and manipulation tasks. Published at IEEE ICRA 2023 (Video).

- **Developed a mobile manipulation system** integrating a PincherX 100 robot arm with a ground rover platform. Implemented grasping and pick-and-place routines using MoveIt and inverse kinematics, with coordinated base mobility for workspace-scale object retrieval.
- **Conducting sim-to-real validation** through ROS/Gazebo simulations and Crazyflie hardware trials to demonstrate safety, robustness, and efficiency of learned policies across varied environmental conditions (Video).

Team Lead — National Center of Robotics & Automation Jan '2020 – May '2022

- **Led a team of 10 engineers** developing autonomous robotic platforms, managing project timelines, coordinating with funding agencies, and mentoring junior engineers on software architecture and best practices.
- **Designed RL-based locomotion control** for snake robot navigation, implementing DQN and PPO policy architectures for autonomous exploration in unstructured environments through learned visuomotor control (Video).
- **Developed autonomous mobile robot** for public-space deployment, integrating real-time crowd density sensing, autonomous navigation, and adaptive behavior policies – validated end-to-end in real-world environments (Video).

Visiting Researcher — Robotics Research Lab, TU Kaiserslautern Jul – Sep '19

- **Created a simulation environment** in Unreal Engine (UE4) and Microsoft AirSim replicating canal-like outdoor scenarios for testing autonomous drone navigation algorithms.
- **Implemented motion planning and collision avoidance algorithms** for autonomous navigation in GPS-denied environments (Video).

Key Projects

- **End-to-End DRL for Collision Avoidance:** Deep RL pipeline in Unreal Engine + Microsoft AirSim for UAV collision-free navigation; sim-to-real transfer validated (95% success rate).
- **System Identification from Data:** Data-driven dynamics modeling pipeline for nano-quadrotors, enabling integration of learned models with classical controllers.
- **Urban Mobility Simulation:** CARLA/ROS multi-agent navigation environment for training and evaluating learned policies under interaction uncertainty.

Research Publications

- **Syed I. Ullah**, et al., "AeroCast: Probabilistic 3D Trajectory Prediction for Non-Cooperative Aerial Obstacles via Transformer-MDN Architecture," *IEEE Tran. on Aerospace and Elec. Sys.*(Under Review, Video).
- **Syed I. Ullah**, et al., "NanoBench: A Multi-Task Benchmark Dataset for Nano-Quadrotor System Identification, Control, and State Estimation," *IEEE Robotics & Automation Letter (RA-L)* (Under Review, Preprint).
- **Syed I. Ullah**, et al., "SynTraG: A Synthetic Trajectory Generator for Non-Cooperative Dynamic Obstacles in UAV Navigation," *Advances in Automation and Robotics Research, Springer, 2026.*(PDF).
- **Syed I. Ullah**, et al., "POF+MADER: Trajectory Planner in Multiagent and Dynamic Environments with Improved Collision Avoidance," *IEEE Access*, vol. 13, 2025. (Video, PDF).
- J. Baca, **Syed I. Ullah**, et al., "Coaxial Modular Aerial System and the Reconfiguration Applications," *IEEE ICRA 2023*, London, UK. (Video, PDF).
- **Syed I. Ullah**, et al., "Autonomous Navigation and Mapping of Snake Robots for Urban Search and Rescue (USAR)," *IEEE ICRAI 2023*. (PDF).
- **Syed I. Ullah**, et al., "Autonomous Navigation and Mapping of Water Channels in a Simulated Environment Using Micro-Aerial Vehicles," *IEEE ICRAI 2023*. (Video, PDF).
- **Syed I. Ullah**, et al., "Motion Planning for a Snake Robot using Double Deep QLearning," *IEEE ICAI 2021(PDF)*.

Professional Certifications & Training

Udacity Nano-Degrees
Robotics Software Engineer, Introduction to Self Driving Cars, Flying Cars & Autonomous Flight Engineer

Coursera Specialization
Mathematics for Machine Learning, Robotics: Computational Motion Planning, Python for Everybody